As AI models become increasingly incorporated into major institutions such as science, education, government, and economics, a collapse toward the most dominant ideas could have severe societal implications. The results of our experiments suggest that differentiation of models could be a significant factor in mitigating such collapse.

2 Background and related work

2.1 Collapse

Model collapse has been theoretically and/or empirically demonstrated for generative models in various settings [1, 17, 31, 56, 70, 73]. The observed effects typically involve some form of performance decay; however, the field has not yet arrived at a general definition of the phenomenon [54]. Our experimental approach for the single model follows Shumailov et al. [56], who first formalized model collapse and revealed it in OPT-125m. Model collapse arises primarily from three sources of error related to training data quality, the model's capacity to represent and accurately learn a given distribution [56]: statistical approximation error, functional expressivity error, and functional approximation error. Prior work show that increasing model and data sizes can amplify collapse [16, 17, 22, 31, 68], thereby contrasting established scaling laws in machine learning. Model collapse is not only a threat to generative models accidentally (or inevitably) trained on artificial internet data but also when considering approaches leveraging synthetic data for efficiency reasons [6, 22, 29]. Very recent work theorizes that model collapse will result in a collapse of knowledge, defined as the continuous narrowing down of ideas, which will lead to broader epistemic problems for our society [49, 64, 68].

2.2 Collapse mitigation

Because of the epistemic risks of collapse, scholars have developed various strategies to prevent such performance decay, or, put differently, to ensure stability. The majority of intervention strategies focus on increasing data quality. This usually involves adding fresh non–model-generated data [e.g., 8, 25, 28], filtering or correcting artificial data [e.g., 3, 17, 22], or diversifying training data [55, 68, 74]. While such methods have been shown to mitigate the problem, in most experimental settings they cannot entirely prevent performance decay. In addition, it is often infeasible technically and practically to constantly source with fresh data or reliably separate real from artificial data [16, 70].

2.3 AI ecosystem

Our real AI ecosystem does not consist of a single self-training model but rather a collection of models trained on both their own outputs and those produced by others. Thus, more closely aligned with the work presented here are recent approaches that seek to understand model collapse at the ecosystem level—for instance, by simulating interactions among AI models via retrieval-augmented generation (RAG) [66], or by sourcing artificial data from multiple language models for training and evaluating a single model [55]. In contrast, our approach focuses on diversity *across* models as the independent variable, involving real (non-simulated) re-training of language models on their collective output.

2.4 Diversity

Epistemic diversity in humans refers to different ways of knowing. It broadly describes the phenomenon that individuals with different background assumptions, shaped by diverse lived experiences, values, and beliefs, can arrive at different interpretations when presented with the same information or data [18, 43, 57].

AI models generate information according to learned probability distributions, which means each model reflects specific assumptions about what counts as relevant, likely, or meaningful knowledge; in other words, a particular epistemology. Epistemic diversity in an AI ecosystem therefore captures the extent to which multiple models, shaped by distinct data sources, architectures, or objectives, may yield divergent outputs when presented with the same input. In recent AI research, epistemic diversity within and across AI models has attracted increased interest. For example, scholars have shown that diverse models collectively make better decisions [38], reduce the risk of failures [44], and enable fairer representations of diverse societies [9, 23, 27, 34, 62].

Originating in ecology, a common method to measure such diversity is the Hill–Shannon Diversity (HSD) [32, 36, 53, 68], which reflects the effective number of distinct, equally common elements (e.g., models, species, or categories) in a population.

3 Approach

We study the progress of model performance in AI ecosystems containing 1, 2, 4, and 16 models, each re-trained on their collective output over 10 iterations. All models begin as identical copies of a pre-trained model but are fine-tuned on equal-sized, non-overlapping subsets of the training data[4]. This segmentation approach enables us to isolate and examine the effect of diversity across models on their performance under fixed model architecture and a fixed per-ecosystem resource budget for data size (number of tokens, N) and compute. We first introduce the primary experimental setting, which demonstrates a clear positive effect of ecosystem diversity on model performance in a controlled, though limited, setting. Then, in Section 3.3, we present variations of this primary setting that aim to approximate more realistic conditions and to examine how the location of the optimum shifts with factors such as model size, data size, incorporation of real data into training, and temperature sampling. We denote the following main variables: D (ecosystem diversity), M (number of models per ecosystem), N (fixed data size per ecosystem, i.e., number of tokens), n (data size per model), T (number of iterations t, i.e., 10), and μ (model perplexity).

Each iteration step involves an evolving **training dataset** used to update model weights, a fixed **generation dataset** used to generate artificial training data for the next iteration, and a fixed **evaluation dataset** used to measure model perplexity (see Figure 2). The evaluation dataset is the same for all models and corresponds to the original Wikitext2 test set. The generation datasets are unique to each model but fixed across all iterations, corresponding to randomly sampled subsets of the original Wikitext2 training data.

In the first iteration step, the training data is identical to the generation dataset for a given model. Thereafter, it evolves in content because it is sourced from the outputs of the previous iteration: At each iteration, the M generated datasets are concatenated, shuffled, and uniformly redistributed to the M models for the next round. Throughout the ten iterations, the training and generation datasets for each model within an ecosystem are equal in size, $n = N/M$. For example, for two models ($D; M = 2$), the sizes of the training and generation datasets are each half of the original size, $n = N/2$ (see Figure 2). In the case of the single-model ecosystem ($D; M = 1$), our approach parallels the methods used in prior work on single-model collapse [e.g., 56].

Translated to the real world, the segmentation approach broadly mirrors domain- or community-specific models [e.g., 13, 21, 60] trained on distinct segments of internet data. While tailored to different communities that might use different inputs, the generated outputs are collectively stored in a single data pool, from which future training samples are drawn.

[4]This approach using non-overlapping train sets is sometimes called bootstrap aggregation (bagging) [15, 26] and has been applied in early work studying performance of model ecosystems [e.g., 39]

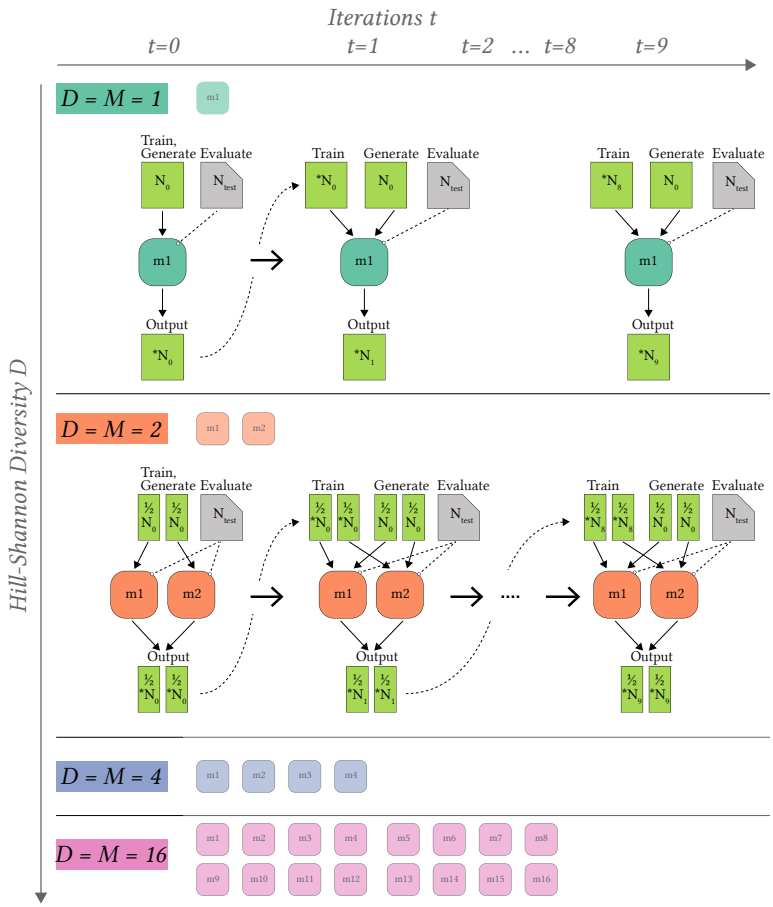

Fig. 2. **Experimental design: manipulating ecosystem diversity and evaluating performance across self-training iterations.** We examine the performance of models m within collections of increasing diversity D when trained on collective output across 10 self-training iterations t. The number of models M equals the Hill-Shannon Diversity D, which is manipulated through segmentation of the training data. At each iteration, we concatenate and shuffle the M generated datasets, and uniformly redistribute to the M models for the next step. This segmentation approach keeps the data size (i.e., number of tokens) per ecosystem N fixed, yielding a per-model sample size of $n = N/M$. The ground-truth dataset used to measure performance, the original Wikitext2 test split, is fixed across all models and iterations.

3.1 Ecosystem diversity D and model performance

Drawing from ecology, we use the HSD to measure ecosystem diversity, that is, diversity across models. The HSD quantifies the effective number of distinct, equally frequent models in a ecosystem [32, 36, 53]. Let M denote the number of models, then we calculate HSD as:

$$D(M) = \exp\left(-\sum_{m=1}^{M} w_m \ln w_m\right) \tag{1}$$

Because of equal weight across models ($w_m = 1/M$), the diversity simplifies to the number of models in a given ecosystem (i.e. richness):